Dynamic Oligopoly and Innovation

Hugo Hopenhayn and Koki Okumura

UCLA

Introduction

- Increase in concentration
- How does concentration affect innovation
- Effect on product market competition and incentives
- Effect on innovation spillovers
  - Business stealing
  - Technology spillovers

Framework

- Quantitative endogenous growth model
- Dynamic R&D involving many oligopolists
  - Existing models: a few firms in each industry (mostly monopolistic competition or duopoly)
  - Our exercise: Public patenting firms in the US ($\simeq$ 1000 firms)
- Quantify inter-firm networks

Networks	Measurement
Product market rivalry	Product proximity (Hoberg and Phillips, 2016): Text analysis of product description in 10k filings
Technology spillover	Technology proximity (Bloom et al., 2013): Patent classification

Literature

- Competition & innovation:
  d'Aspremont and Jacquemin (1988); Aghion et al. (2001, 2005); Acemoglu and Akcigit (2012); Aghion et al. (2013); Bloom et al. (2013); Lopez and Vives (2019); Peters (2020); Akcigit and Ates (2021, 2023); Liu et al. (2022); Cavenaile et al. (2023); Anton et al. (2023, 2024); Kini et al. (2024)
  Quantitative endogenous growth model matched to firm-(pair-)level data
- Hedonic demand / empirical IO:
  Lancaster (1966); Rosen (1974); Epple (1987); Berry et al. (1995); Nevo (2001); Pellegrino (2024); Ederer and Pellegrino (2024)
  Dynamic general equilibrium / R&D
- Oligopoly / market power:
  Neary (2003); Atkeson and Burstein (2008); Gutierrez and Philippon (2017); Autor et al. (2020); Baqaee and Farhi (2020); De Loecker et al. (2020); Azar and Vives (2021); Edmond et al. (2023)

Generalized Hedonic-Linear Demand (Pellegrino, 2024) GHL

- $i \in \{1, 2, ..., n\}$: firms / products
- $\mathbf{q}_t = [q_1, q_2, ...q_n]$: vector of quantities
- Linear quadratic aggregator:

$$Y_t = \mathbf{q}_t^T \mathbf{b}_t - \frac{1}{2}\mathbf{q}_t^T \Sigma \mathbf{q}_t$$

- Linear inverse demand:

$$\mathbf{p}_t = \mathbf{b}_t - \Sigma \mathbf{q}_t$$

- $\mathbf{b}_t$: vector of product qualities
- Σ: matrix of product substitutability

Production Technology and Technology Choice

- Production technology: $q_{i,t} = a_{i,t} l_{i,t}$
- Each firm has their knowledge capital $z_{i,t}$
- Each firm allocates their knowledge capital to improve product quality and labor productivity:

$$\zeta a_{i,t} + b_{i,t} = z_{i,t}$$

Law of Motion of Knowledge Capital

$$d\mathbf{z}_t = \left(\underbrace{\Omega \mathbf{z}_t}_{\text{technology spillover}} + \underbrace{\mu \mathbf{x}_t}_{\text{R\&D}} \right) dt + \underbrace{\gamma \mathbf{z}_t dW_t}_{\text{shocks}}$$

- $\Omega = [\omega_{ij}]$: technology spillover matrix
- $x_{i,t} = \sqrt{d_{i,t}}$
  - $d_{i,t}$: R&D cost in terms of final good
  - Innovation elasticity is 0.5
- μ, γ: positive scalars

Market Clearing and Preference

- Inelastic labor supply:

$$L = \sum_i l_{i,t}$$

- Final good market clearing:

$$C_t + \underbrace{\sum_i d_{i,t}}_{\text{R\&D cost}} = Y_t$$

- Risk neutral representative household:

$$\max \mathbb{E}_t \left[\int_t^\infty \exp\left(-\rho s\right) C_s ds \right]$$

Cournot-Nash Equilibrium

- Firm i's gross profit before subtracting dynamic R&D cost:

$$\pi_{i,t} = p_{i,t} q_{i,t} - w_t l_{i,t} = \left(b_{i,t} - \frac{1}{a_{i,t}} w_t - \sum_j \sigma_{ij} q_{j,t} \right) q_{i,t} \quad \text{where} \quad \zeta a_{i,t} + b_{i,t} = z_{i,t}$$

- Given w_t, $z_{i,t}$, and $\{q_{j,t}\}_{j \neq i}$, firm i chooses $a_{i,t}$, $b_{i,t}$, and $q_{i,t}$ to maximize $\pi_{i,t}$

Cournot-Nash Equilibrium

- Firm i's gross profit before subtracting dynamic R&D cost:

$$\pi_{i,t} = p_{i,t} q_{i,t} - w_t l_{i,t} = \left(b_{i,t} - \frac{1}{a_{i,t}} w_t - \sum_j \sigma_{ij} q_{j,t} \right) q_{i,t} \quad \text{where} \quad \zeta a_{i,t} + b_{i,t} = z_{i,t}$$

- Given w_t, $z_{i,t}$, and $\{q_{j,t}\}_{j \neq i}$, firm i chooses $a_{i,t}$, $b_{i,t}$, and $q_{i,t}$ to maximize $\pi_{i,t}$
- Quantity is a linear function of knowledge capital:

$$\mathbf{q}_t = \left\{ \underbrace{2\frac{\zeta}{L} J}_{\text{labor cost}} + \underbrace{\Sigma}_{\text{substitutability}} + I \right\}^{-1} \mathbf{z}_t$$

  - J: matrix with all elements equal to 1 / I: identity matrix
- Gross profits are expressed in quadratic form: $\pi_{i,t} = \mathbf{z}_t^T Q^i \mathbf{z}_t$ Q

Linear-Quadratic Differential Game

- Given other players' strategy $\{x_{j,t}\}_{j\neq i, t\geq 0}$, firm i chooses R&D effort $\{x_{i,t}\}_{t\geq 0}$ to maximize

$$\max_{\{x_{i,t}\}_{t\geq 0}} \quad V^i(\mathbf{z}_0) \equiv \mathbb{E}_0\left[\int_0^\infty \exp(-\rho t)\left\{\pi_{i,t} - x_{i,t}^2\right\} dt\right]$$

subject to $d\mathbf{z}_t = (\Omega \mathbf{z}_t + \mu \mathbf{x}_t)\, dt + \gamma \mathbf{z}_t dW_t$

- Firm i's HJB equation:

$$\rho V^i(\mathbf{z}) = \max_{x_i}\left\{\mathbf{z}^T Q^i \mathbf{z} - x_j^2 + V_{\mathbf{z}}^i(\mathbf{z})\left[\Omega \mathbf{z} + \mu \mathbf{x}\right] + \frac{\gamma^2}{2}\mathbf{z}^T V_{\mathbf{zz}}^i(\mathbf{z})\,\mathbf{z}\right\}$$

HJB Equations $\Longrightarrow$ Riccati Equations

- Guess and verify $V^i(\mathbf{z}) = \mathbf{z}^T X^i \mathbf{z}$ (for any $\mathbf{z}$)
- X^i is the solution of stacked algebraic Riccati equations Riccati
- All public patenting firms in the US in our dataset $\simeq$1000 firms $\Longrightarrow$
  - $1000^3 = 1$ billion undetermined coefficients

Oligopolistic Schumpeterian growth model	Computation time (n firms)	Number of firms	Space of productivity
Cavenaile et al. (2023)	$O(2^n)$	4	6 grids
Our model	$O(n^4)$	Around 1000	Continuous

BGP Equilibrium

- R&D strategy: $x_{i,t} = \left(\mu X_i^i\right)^T \mathbf{z}_t$ where X_i^i is the i th column of X^i
- The law of motion is rewritten as $d\mathbf{z}_t = \Phi \mathbf{z}_t dt + \gamma \mathbf{z}_t dW_t$ where

$$\underbrace{\Phi}_{\text{technology transition matrix}} \equiv \underbrace{\Omega}_{\text{spillover}} + \underbrace{\mu^2 \begin{bmatrix} X_1^1 & \cdots & X_n^n \end{bmatrix}^T}_{\text{R\&D}}$$

Theorem

Consider the deterministic economy ($\gamma = 0$). If the technology transition matrix Φ is irreducible,
(i) there exists a largest positive eigenvalue of Φ, g, and associated strictly positive eigenvector, $\mathbf{z}^*$
(ii) there exists a globally stable BGP such that the knowledge capital growth rate of all firms is g and knowledge capital distribution is a scalar multiple of $\mathbf{z}^*$

- Proof: Perron–Frobenius Theorem
- "Φ is irreducible" $\Longleftrightarrow$ "All firms are directly or indirectly connected technologically"

BGP in Deterministic Economy

- In deterministic BGP, $\mathbf{a}_t$, $\mathbf{b}_t$, $\mathbf{z}_t$, and $\mathbf{q}_t$ grow at the same rate

Technology Choice	$\zeta a_{i,t} + b_{i,t} = z_{i,t}$
Linear Production Technology	$q_{i,t} = a_{i,t} l_{i,t}$
Inelastic Labor Supply	$L = \sum_i l_{i,t}$

- Output is expressed in quadratic form: Equilibrium Summary

$$Y_t = \mathbf{q}_t^T \mathbf{b}_t - \frac{1}{2}\mathbf{q}_t^T \Sigma \mathbf{q}_t = \mathbf{z}_t^T Q \mathbf{z}_t$$

Aggregation

- Apply Ito's lemma to $\log Y_t = \log\left(\mathbf{z}_t^T Q \mathbf{z}_t\right)$ and $d\mathbf{z}_t = \Phi \mathbf{z}_t dt + \gamma \mathbf{z}_t dW_t \Longrightarrow g_t$ Decomposition
- Expected utility is expressed in quadratic form: X

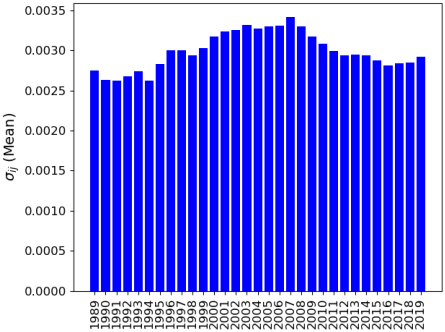
$$\mathbb{E}_t\left[\int_t^\infty \exp\left(-\rho s\right) C_s ds \middle| \mathbf{z}_t\right] = \mathbf{z}_t^T X \mathbf{z}_t$$

- Solve the equilibrium once $\Longrightarrow$ Can compute expected growth and utility for any $\mathbf{z}_t$

Product Substitutability

$$\underbrace{\sigma_{ij}}_{\text{substitutabiliy}} = \alpha \times \text{product similarity b/w } i \text{ and } j \quad (i \neq j)$$

- Hoberg and Phillips (2016) estimates product similarity using product descriptions in 10-K
- Pellegrino (2024) estimates α to align with the cross-price elasticity of demand



Technological Proximity

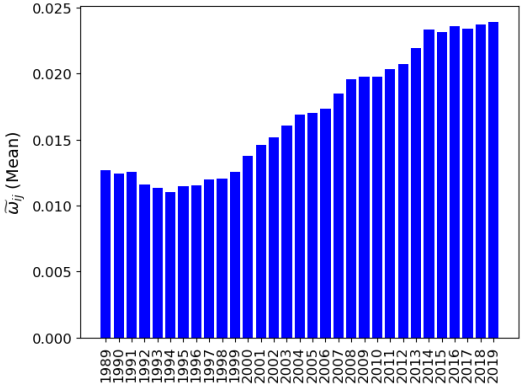
- T_i: the vector representing the share of patents held by firm i in each technology class
  - (i) subclass-level ($\approx$ 600) or (ii) group-level ($\approx$ 4000) classifications
- Jaffe (1986) technology proximity measure:

$$\tilde{\omega}_{ij} = \frac{T_i T_j'}{\left(T_i T_i'\right)^{1/2} \left(T_j T_j'\right)^{1/2}}$$

  - Assumption: Spillovers exist only within the same technology classes
- Mahalanobis measure for robustness (Bloom et al., 2013; Arora et al., 2021)
  - Accounts for spillovers between technology classes
  - Based on frequency with which patents are taken out in different classes within firm

Technological Proximity

- Merge USPTO data with Compustat firms using DISCERN 2 dataset (Arora et al., 2024)
- Jaffe measure, Group-level patent classification, Stack for 5 years



Initial Distribution of Knowledge Capital

1. Gross profit (before subtracting R&D cost) $\pi_{i,t} \Longleftarrow$ "Revenue" $-$ "Costs of goods sold"
   - Deflated by GDP deflator for inter-temporal comparison
2. Quantity $\mathbf{q}_t \Longleftarrow \pi_{i,t} = q_{i,t}^2$
3. $\zeta / L \Longleftarrow$ match the cost share of the sample firms
4. Knowledge capital $\mathbf{z}_t \Longleftarrow \mathbf{q}_t = \left\{ 2\frac{\zeta}{L} J + \Sigma + I \right\}^{-1} \mathbf{z}_t$

Technology Spillover $\Omega = \beta \times$ Technology Proximity $\tilde{\Omega}$

$$z_{i,t+1} - z_{i,t} = \beta \sum_{j \neq i} \tilde{\omega}_{ij,t} z_{j,t} + \text{Year FE}_t + \varepsilon_{i,t}$$

	(1)	(2)	(3)
Dependent Variable:	$z_{i,t+1} - z_{i,t}$	$z_{i,t+1} - z_{i,t}$	$z_{i,t+1} - z_{i,t}$
$\sum_{j \neq i} \tilde{\omega}_{ij,t} z_{j,t}$	0.000103***	0.000082***	0.000081***
	(0.000017)	(0.000018)	(0.000021)
$\sqrt{\text{R\&D Expenditure}}$		.046**	
		(0.012)	
Year Fixed Effects	✓	✓	✓
IV			✓
IV 1st Stage F-statistics			9661
Patent Classification	Group	Group	Group
Technological Proximity	Jaffe	Jaffe	Jaffe
No. observations	35,254	32,465	31,385

SEs clustered by years and 4-digit naics industries are reported in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. First Stage

- IV: User cost of R&D, driven by state-level tax variations (Wilson, 2009; Bloom et al., 2013)

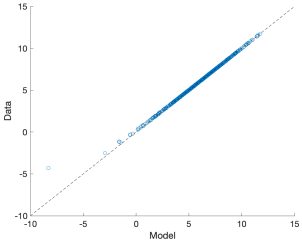
Identification: Summary

- Publicly available data + Compustat

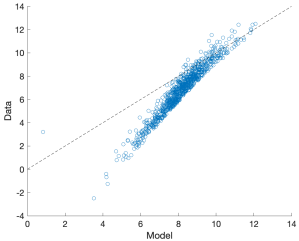
Notation	Description	Value	Source
$\Psi^T\Psi$	Product proximity		Form 10-K, Hoberg and Phillips (2016)
$\widetilde{\Omega}$	Technological proximity		USPTO, Patent classification
α	Product proximity $\rightarrow$ Substitutability	.12	Pellegrino (2024)
β	Technological proximity $\rightarrow$ Spillover	.000081	Estimate the law of motion
γ	St.d. of idiosyncratic shocks	.030	Estimate the law of motion
ζ/L	Labor augmentation efficiency	.0039	Compustat, Cost of goods sold
ρ	Discount rate	.10	
μ	R&D efficiency	.05	1.6% economic growth rate

Goodness of Fit b/w Model and Data

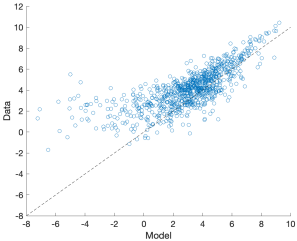
- Model (x-axis) vs Data (y-axis) : firm-level profits (targeted), sales and R&D expenditures



(a) Log of Profit (Targeted)

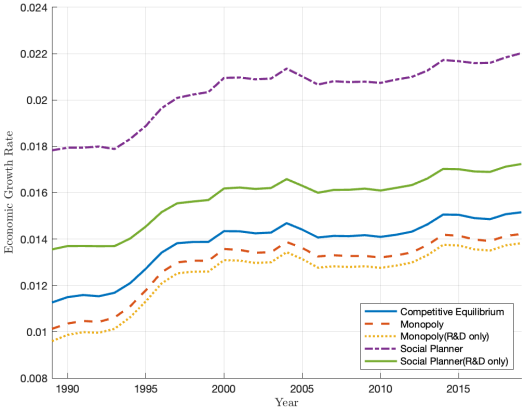


(b) Log of Sales

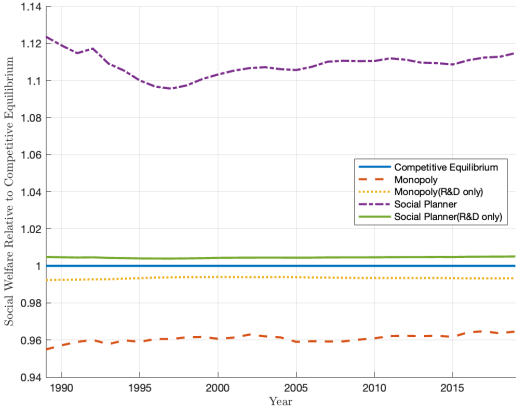


(c) Log of R&D Expenditure

Growth Rates: Monopoly vs Competitive Equilibrium vs Planner

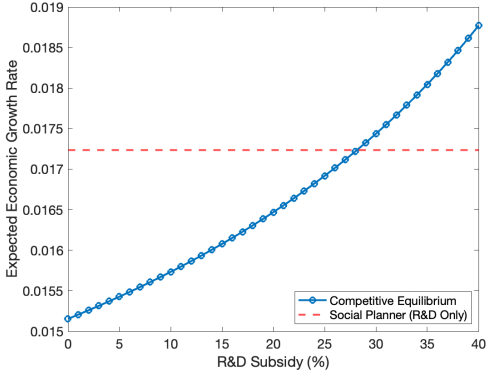


Social Welfare: Monopoly vs Competitive Equilibrium vs Planner

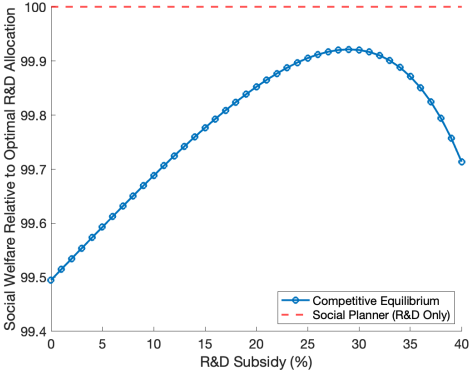


Uniform R&D Subsidy

Social Optimum



(a) Expected Growth



(b) Expected Social Welfare

- Optimal rate is $s = 28\%$, which increases g by 0.23 pp
- CE Welfare loss relative to optimal R&D allocation is reduced to 0.1% (Initially 0.5%)

Conclusion

Generalized Hedonic-Linear Demand (Pellegrino, 2024)

- $i \in \{1, 2, \ldots, n\}$: firms / products
- 1 unit of product i provides
  - 1 unit of idiosyncratic characteristic $k \in \{1, 2, \ldots, n\}$
  - $\psi_{k,i}$ unit of shared characteristic $k \in \{n+1, n+2, \ldots, n+n_k\}$ where $\sum_k \psi_{k,i}^2 = 1$
- Aggregate each characteristic:

$$y_{k,t} = \begin{cases} q_{k,t} & k = 1, 2, \ldots, n \\ \sum_i \psi_{k,i} q_{i,t} & k = n+1, n+2, \ldots, n+n_k \end{cases}$$

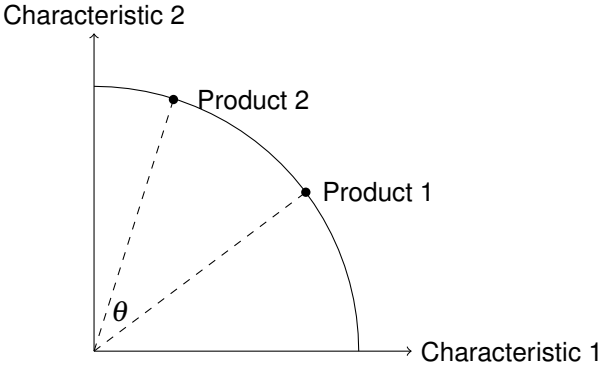
- Linear quadratic aggregator over characteristics:

$$Y_t = (1-\alpha) \sum_{k=1}^{n} \left(\underbrace{\hat{b}_{k,t} y_{k,t} - \frac{1}{2} y_{k,t}^2}_{\text{idiosyncratic characteristic}} \right) + \alpha \sum_{k=n+1}^{n+n_k} \left(\underbrace{\hat{b}_{k,t} y_{k,t} - \frac{1}{2} y_{k,t}^2}_{\text{shared characteristic}} \right)$$

Product Substitutability $\propto$ Product Proximity (Pellegrino, 2024)

- $\Sigma = [\sigma_{ij}]$ is estimated using product proximity measure
- Product proximity is based on text analysis of product description in 10k filings

$$\sigma_{ij} = \alpha \times \text{product cosine similarity b/w } i \text{ and } j \quad (i \neq j)$$



Generalized Hedonic-Linear Demand (Pellegrino, 2024)

- Quality:

$$b_i = (1-\alpha)\,\hat{b}_i + \alpha \sum_{k=n+1}^{n+n_k} \psi_k \hat{b}_k$$

- Inverse demand:

$$\frac{\mathbf{p}}{P} = \mathbf{b} - \Sigma \mathbf{q}$$

- Inverse cross price elasticity of demand:

$$\frac{\partial \log p_i}{\partial \log q_j} = -\frac{q_j}{p_i} \cdot \sigma_{ij}$$

- Cross price elasticity of demand:

$$\frac{\partial \log q_i}{\partial \log p_j} = -\frac{p_j}{q_i} (\Sigma^{-1})_{ij}$$

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Static Profits

- Real gross profit: $\frac{\pi_{i,t}}{P_t} = q_{i,t}^2$
- Firms choose labor productivity and product quality: $\zeta a_{i,t} = \sqrt{\zeta \frac{w_t}{P_t}}$, $b_{i,t} = z_{i,t} - \sqrt{\zeta \frac{w_t}{P_t}}$
- Labor market clearing: $L = \sum_i \frac{q_{i,t}}{a_{i,t}} \Longrightarrow \sqrt{\zeta \frac{w_t}{P_t}} = \frac{\zeta}{L} \sum_i q_{i,t}$
- $\mathbf{q}_t = N\mathbf{z}_t$ where $N \equiv \left\{ 2\frac{\zeta}{L} J + \Sigma + I \right\}^{-1}$
- N_i: the i th row of N
- Ownership weighted profit:

$$\frac{\pi_{j,t}}{P_t} = q_{i,t}^2 = \mathbf{z}_t^T Q^i \mathbf{z}_t$$

where

$$Q^i = N_i^T N_i$$

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Riccati Equations

- $V^i(\mathbf{z}) = \mathbf{z}^T X^i \mathbf{z}$ where X^i is the solution of the stacked Riccati equation

$$0 = Q^i - \mu^2 \sum_j \kappa_{ij} X_j^j \left(X_j^j\right)^T + \left(\Phi - \frac{1}{2}\left(\rho - \gamma^2\right) I\right)^T X^i + X^i \left(\Phi - \frac{1}{2}\left(\rho - \gamma^2\right) I\right)$$

  - $X_i^j \equiv$ the i th column of X^i
  - $\Phi \equiv \Omega + \mu^2 \begin{bmatrix} X_1^1 & \cdots & X_n^n \end{bmatrix}^T$
- Algorithm: Given $\begin{bmatrix} X_\tau^1 & \cdots & X_\tau^n \end{bmatrix}$, update $\begin{bmatrix} X_{\tau-\Delta}^1 & \cdots & X_{\tau-\Delta}^n \end{bmatrix}$ by

$$-\frac{X_\tau^i - X_{\tau-\Delta}^i}{\Delta} = Q^i - \mu^2 \sum_j \kappa_{ij} X_{j,\tau}^j \left(X_{j,\tau}^j\right)^T + \left(\Phi_\tau - \frac{1}{2}\left(\rho - \gamma^2\right) I\right)^T X_\tau^i + X_\tau^i \left(\Phi_\tau - \frac{1}{2}\left(\rho - \gamma^2\right) I\right)$$

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Summary of Equilibrium

Description	Expression
Production strategy	$\mathbf{q}_t = N\mathbf{z}_t$
R&D strategy	$\mathbf{x}_t = \mu \tilde{X} \mathbf{z}_t$
Law of motion	$d\mathbf{z}_t = (\Omega \mathbf{z}_t + \mu \mathbf{x}_t)\, dt + \gamma \mathbf{z}_t dW_t$
Profit of final producers	$\Pi_t^F / P_t = \mathbf{q}_t^T \left(\frac{1}{2}\Sigma\right) \mathbf{q}_t$
Total operating profit of firms	$\Pi_t / P_t = \mathbf{q}_t^T \mathbf{q}_t$
Labor income	$w_t L / P_t = \mathbf{q}_t^T \left(\frac{\zeta}{L} J\right) \mathbf{q}_t$
Output	$Y_t = \mathbf{q}_t^T \left(\frac{\zeta}{L} J + \frac{1}{2}\Sigma + I\right) \mathbf{q}_t$
Consumption	$C_t = Y_t - \mathbf{x}_t^T \mathbf{x}_t$

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Output and Expected Utility

- Output: $Y_t = \mathbf{q}_t^T Q \mathbf{q}_t$ where

$$Q = \frac{\zeta}{L} J + \frac{1}{2}\Sigma + I$$

- Expected utility:

$$V(\mathbf{z}_t) \equiv \mathbb{E}_t \left[\int_t^\infty \exp(-\rho s) C_s ds \middle| \mathbf{z}_t \right] = \mathbf{z}_t^T X \mathbf{z}_t$$

where X is the solution of the Lyapunov equation (obtained from households' HJB equation):

$$0 = Q - \mu^2 \tilde{X}^T \tilde{X} + X \left(\Phi - \frac{1}{2}(\rho - \gamma^2) I \right) + \left(\Phi - \frac{1}{2}(\rho - \gamma^2) I \right)^T X$$

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Social Optimum

- Static optimal allocation: $\mathbf{q}_t^* = N^* \mathbf{z}_t$ where $N^* \equiv \left\{ 2\frac{\zeta}{L} J + \Sigma \right\}^{-1}$
- Optimal output: $Y_t^* = \mathbf{z}_t^T Q^* \mathbf{z}_t$ where $Q^* = \frac{1}{2} N^*$
- Optimal expected utility:

$$V^*(\mathbf{z}_t) \equiv \mathbb{E}_t \left[\int_t^\infty \exp(-\rho s) C_s ds \middle| \mathbf{z}_t \right] = \mathbf{z}_t^T X^* \mathbf{z}_t,$$

where X^* is the solution of the Riccati equation (obtained from planner's HJB equation):

$$0 = Q^* - \mu^2 (X^*)^2 + X^* \left(\Phi^* - \frac{1}{2} (\rho - \gamma^2) I \right) + \left(\Phi^* - \frac{1}{2} (\rho - \gamma^2) I \right) X^*$$

- Optimal R&D: $\mathbf{x}_t^* = \mu X^* \mathbf{z}_t$
- Optimal technology transition matrix: $\Phi^* = \Omega + \mu^2 X^*$

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Stochastic Process of Output

- Applying Itô's lemma,

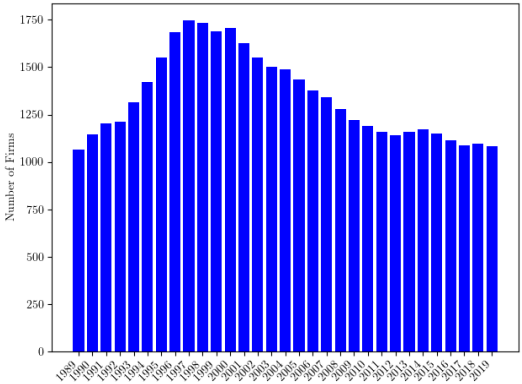
$$d\log Y_t = \left[\frac{\mathbf{z}_t^T\left(Q\Phi + \Phi^T Q\right)\mathbf{z}_t}{Y_t} + \gamma^2\left\{\frac{\sum_i z_{i,t}^2 Q_{ii}}{Y_t} - \frac{2\mathbf{z}_t^T Q\,\mathrm{diag}\left(\mathbf{z}_t^2\right)Q\mathbf{z}_t}{Y_t^2}\right\}\right]dt + \frac{2\gamma\mathbf{z}_t^T Q\,\mathrm{diag}\left(\mathbf{z}_t\right)}{Y_t}dW_t$$

where $Y_t = \mathbf{z}_t^T Q\mathbf{z}_t$ and $\Phi = \Omega + \mu^2\widetilde{X}$

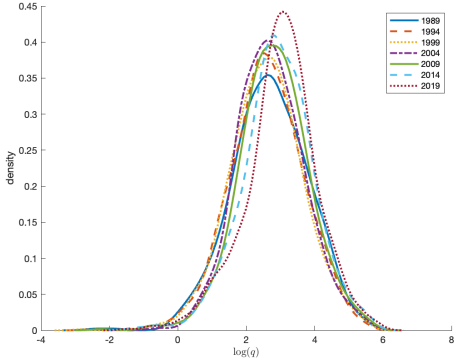
Description	Expression
Tech Spillover	$\mathbf{z}_t^T\left(Q\Omega + \Omega Q\right)\mathbf{z}_t / Y_t$
R&D	$\mu^2\mathbf{z}_t^T\left(Q\widetilde{X} + \widetilde{X}^T Q\right)\mathbf{z}_t / Y_t$
Ito	$\gamma^2\left\{\sum_i z_{i,t}^2 Q_{ii}/Y_t - 2\mathbf{z}_t^T Q\mathrm{diag}\left(\mathbf{z}_t^2\right)Q\mathbf{z}_t/Y_t^2\right\}$
Total	$\mathbb{E}\left[d\log Y_t\right]$

Back

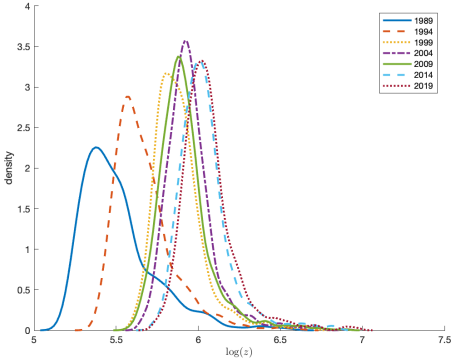
Number of Sample Firms



Distributions of Estimated Knowledge Capital and Quantity



(a) Log of Quantity



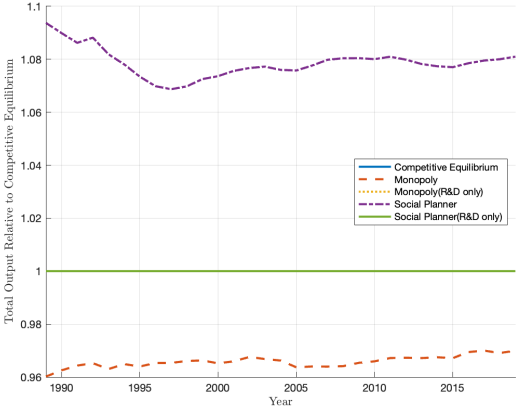
(b) Log of Knowledge Capital

First Stage Back

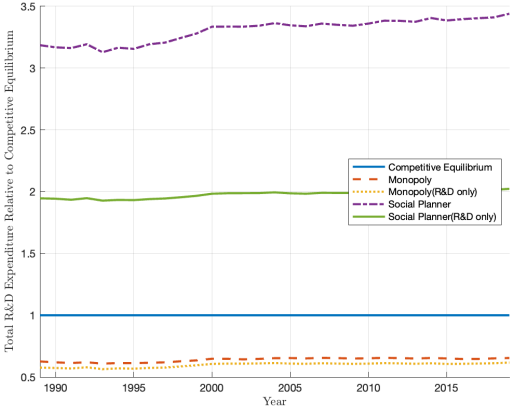
	(1)
Dependent Variable:	$z_{i,t}$
User cost of R&D	-10.55^{***}
	(3.62)
Year Fixed Effects	✓
No. observations	30,046

SEs clustered by years and 4-digit naics industries are reported in parentheses.

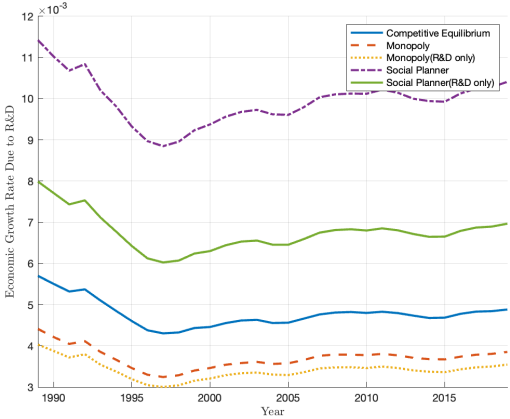
Output


1.1
1.08
1.06
1.04
1.02
1
0.98
0.96
Total Output Relative to Competitive Equilibrium
1990
1995
2000
2005
2010
2015
Year
Competitive Equilibrium
Monopoly
Monopoly(R&D only)
Social Planner
Social Planner(R&D only)

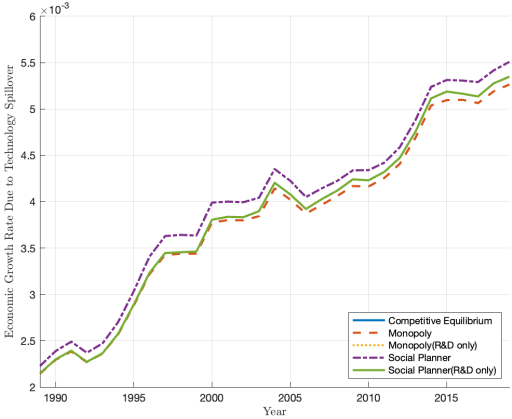
R&D Expenditure



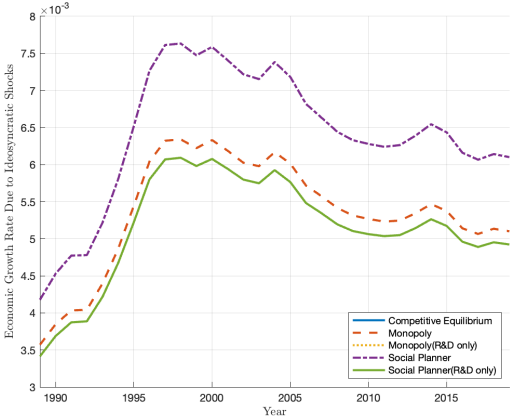
Growth Rate Due to R&D



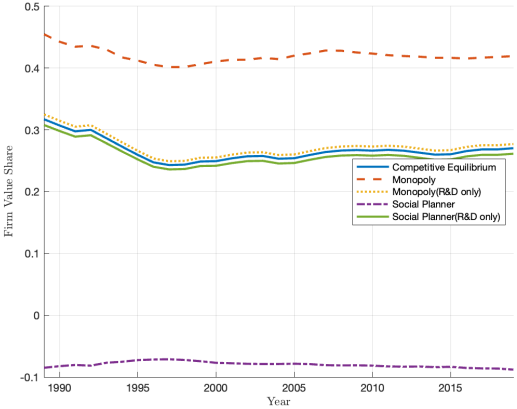
Growth Rate Due to Technology Spillover



Growth Rate Due to Ito Process



Firm Value Share



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